

Climate Change Twitter Dataset

End-to-End SQL & Power BI Analysis

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15.8M Total Tweets	2006–2019 Date Range	0.0025 Avg Sentiment	10 Topics Covered
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1. Introduction & Dataset Overview

This report presents an end-to-end analysis of the Climate Change Twitter Dataset (Effrosynidis, 2022), conducted using PostgreSQL for all data processing and Power BI for visualisation. The dataset contains 15.8 million tweets spanning June 2006 to October 2019 — the documented range of 2008 to 2022 was found to be inaccurate on both ends during investigation.

The analysis was structured around two questions from leadership:

- How have public sentiment and stance on climate change shifted across 13 years of Twitter discourse?
- Which topics and regions are driving the most divisive or aggressive conversations?

Each tweet carries seven dimensions: timestamp, geolocation, topic (LDA), sentiment score (-1 to +1), stance (believer/denier/neutral), author gender, temperature deviation, and aggressiveness label — all generated by ML models including BERT, LSTM, SVM, and VADER.

2. Data Quality, Cleaning & Transformation

2.1 Investigation Findings

Before any cleaning was applied, every column was investigated systematically. The goal was to let the data indicate what needed fixing rather than assume issues upfront — an approach that prevented several unnecessary operations.

Check	Finding	Decision
Duplicates	Zero duplicate tweet IDs	No action needed
Missing coordinates	10.48M rows (66%) have no lng/lat	Retained — inherent to Twitter's optional location tagging
Missing temperature	Same 10.48M rows — derived from location	Retained — linked limitation
Sentiment range	Min -0.994, Max 0.991	Within bounds, no action needed
Coordinate validity	All within valid geographic bounds	No action needed
Suspicious coordinates	Two London pairs flagged as possible defaults	Retained after frequency analysis confirmed normal city volume
Topic typo	'Intervantion' misspelled in source data	Corrected to 'Intervention' during cleaning
Categorical casing	Mixed case across stance, gender, aggressiveness	Standardised to lowercase

2.2 Cleaning & Continent Assignment

A separate clean table was created from the raw data, preserving the original at all times. Beyond the typo fix and casing standardisation, a continent column was derived using coordinate boundary ranges in SQL — the only viable approach without Python or an external geocoding API.

During verification, Central American countries were falling through the initial North America boundary (set at latitude 15). Extending it to latitude 7 correctly captured Costa Rica, Honduras, and El Salvador. The final geotagged distribution: North America 64.9%, Europe 19.4%, Oceania 6.2%, Asia 4.9%, Africa 2.6%, South America 1.3%. All regional findings are limited to the 34% of tweets with valid coordinates.

3. Descriptive Analytics

3.1 Volume & Stance

Tweet volume grew substantially over the period, with 2018 accounting for roughly 40% of all tweets (6.26M). The 2009 spike to 489,000 tweets aligns with the Copenhagen Climate Summit. Volume growth accelerates from 2015 onwards, coinciding with the Paris Agreement and increased political polarisation.

Believers dominate throughout at 71.52%, with deniers making up only 7.55%. Crucially, denier average sentiment is -0.232 — significantly more negative than believers (0.017) and neutrals (0.039). This pattern holds consistently across all years.

3.2 Topics & Aggressiveness

Politics is the most aggressive topic at 43.4% despite not being the largest by volume. Significance of Pollution Awareness Events is the least aggressive at 15.95%, suggesting awareness-driven content attracts calmer discourse than politically charged topics. Overall, 28.67% of all tweets are classified as aggressive.

Gender plays no meaningful role in aggressiveness — male and female users show nearly identical aggressive rates at 27.9% and 27.8% respectively, despite males accounting for roughly twice the tweet volume.

4. Diagnostic Analytics

4.1 Stance and Aggressiveness

Deniers are 15 percentage points more aggressive than believers — a gap that is consistent across every year and every region in the dataset:

Stance	Tweets	Aggressive Rate	Avg Sentiment
Denier	1,191,386	42.65%	-0.232
Neutral	3,305,601	28.21%	0.039
Believer	11,292,424	27.33%	0.017

Denier sentiment is negative in every single year from 2006 to 2019, never crossing zero. Believer sentiment trends positively from 2015 onwards — the two groups are diverging rather than converging over time.

4.2 Regional & Topic Patterns

North America is the most aggressive region at 31.78%, driven primarily by Politics (45.8%) and Donald Trump versus Science (43.4%). Asia is the least aggressive at 17.93%. Africa and Asia show the most positive average sentiment despite the smallest tweet volumes.

Politics is the most aggressive topic in every continent except Oceania, where Donald Trump versus Science takes the top spot. This cross-regional consistency suggests the political framing of climate discourse — not location — is the primary driver of aggression.

4.3 Temperature and Aggressiveness

Temperature deviation shows no meaningful relationship with aggressiveness. Unusually cold conditions (30.81%), normal range (28.70%), and unusually hot conditions (28.53%) all sit within 2.3 percentage points of each other. External weather does not appear to drive aggressive climate discourse online — political and ideological framing does.

5. Dashboard, Recommendations & Conclusions

5.1 Dashboard Architecture

The Power BI dashboard connects to two PostgreSQL views. The master view (vw_master_climate_metrics) groups all 15.8M rows by seven dimensions, reducing the data to ~103,000 rows for fast cross-filtering. A spatial binning view (vw_map_grid) was built to plot coordinates at an 11km grid resolution, but was replaced in the final dashboard with continent-level aggregations due to rendering performance issues in the published environment.

The dashboard is structured across three pages: Overview (dataset snapshot and distributions), Sentiment & Stance Over Time (answering Q1), and Topics, Regional & Aggression (answering Q2). All pages share consistent slicers for Year, Continent, Topic, and Stance, enabling full cross-filtering interactivity.

Published Dashboard:

<https://app.powerbi.com/view?r=eyJrIjoic4NjIINDU0YzJhOS00NTNLTGwYzItNGE5ZGMwNjkzMzQxIiwidCI6ImMwZmMxNTU3LTEzMzktNDZkZC05MDk1LWExZjM0ZDdiZmI3ZiJ9>

5.2 Recommendations

- Communication targeting denier communities should anticipate hostile engagement. The aggressiveness gap is structural and consistent — not situational.
- Political framing consistently produces the most aggressive responses across all regions. Awareness-driven messaging produces the least. This has direct implications for how policy communications should be structured.
- North America and Oceania are the regions where aggressive discourse is most concentrated and should be prioritised in targeted communication strategies.
- Believer sentiment has trended positive since 2015 — an opportunity to build on growing confidence in the pro-climate community rather than defaulting to defensive framing.

5.3 Limitations & Conclusions

Twitter demographics skew younger, urban, and English-speaking — these findings reflect Twitter discourse, not global public opinion. The dataset ends in October 2019, predating significant climate events including COP26. Sentiment and aggressiveness scores carry the inherent margin of error of the ML models that generated them.

Despite these limitations, the dataset is large enough and temporally broad enough to surface consistent patterns that are unlikely to be noise. The clearest conclusion is that climate denial on Twitter is not just a minority position — it is a structurally more aggressive and more negative one, and has been for over a decade.

5.4 Reference

Effrosynidis, D. (2022). The Climate Change Twitter Dataset. Mendeley Data, V2.
<https://doi.org/10.17632/mw8yd7z9wc.2>

Associated paper: <https://doi.org/10.1016/j.eswa.2022.117541>

End of Report